GitHub

GitHub Copilot fundamentals




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Agenda

Copilot essentials

Your first steps with GitHub Copilot

Copilot features

A quick tour into GitHub Copilot’s features



SESSION LEARNING OBJECTIVES

GitHub Copilot fundamentals

- Get started with GitHub Copilot and use its core features
- Optimize your GitHub Copilot experience using best practices and techniques

GITHUB DOCS

An AI-powered code completion tool that helps developers write code faster.





Who is GitHub Copilot for?

Senior Engineers

Copilot accelerates boilerplate code generation, leverages design patterns, and simplifies complex problem-solving.

Junior Engineers

Copilot speeds up learning, facilitates as a pair programmer, and mentors.



Copilot vs Copilot Chat

GitHub Copilot

- Direct code writing
- Seamless IDE integration
- Solo development

GitHub Copilot Chat

- In-depth assistance
- Learning and teaching
- Collaborative scenarios

Limitations

Training data impacts suggestions

Models are trained based on a certain date

GitHub Copilot is not a compiler

Always test out the suggestions given

GitHub Copilot can’t read your mind.

Copilot needs your guidance and vision





“

I’m powered by AI, so surprises and mistakes are possible. Make sure to verify any generated code or suggestions, and share feedback so that we can learn and improve.

GitHub Copilot

Prerequisites

What you need



GitHub account



Copilot license



Supported IDE

Compatibility



Visual Studio

Code completions and GitHub Copilot Chat



VS Code

Code completions and GitHub Copilot Chat



JetBrains Suite

Code completions and GitHub Copilot Chat



Xcode

Code completions and GitHub Copilot Chat



OpenCode

Code completions and GitHub Copilot Chat



Eclipse

Code completions and GitHub Copilot Chat



Vim/Neovim

Code completions



Windows Terminal

Code completions and GitHub Copilot Chat



GitHub.com, GitHub Mobile and Copilot app

GitHub Copilot Chat



And more!

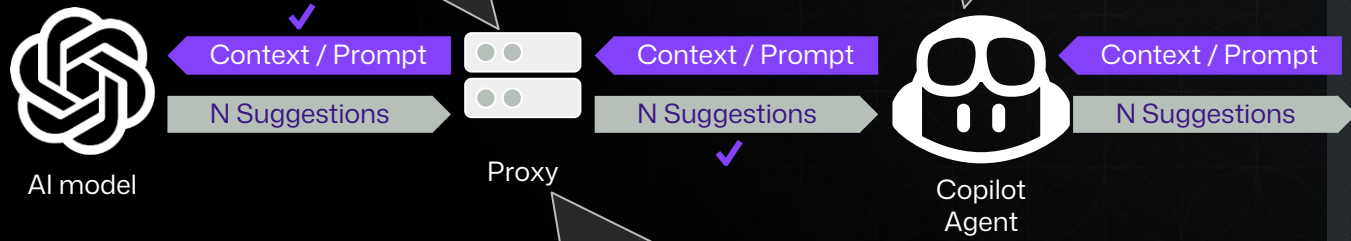
New clients added all the time!

How GitHub Copilot fulfills a request

Apply several **pre-model filters**

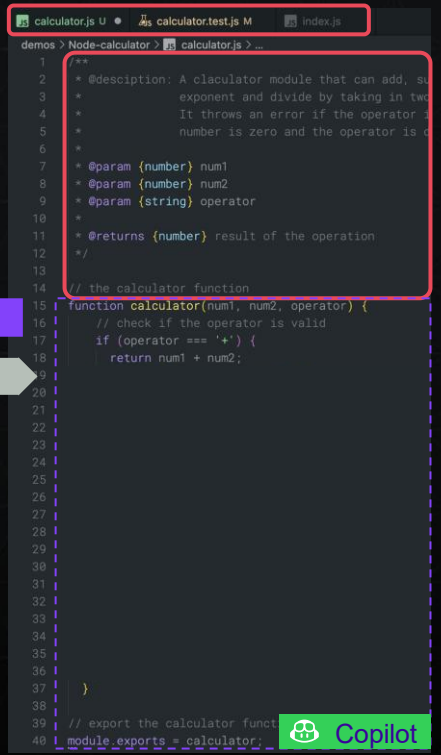
- Test for **toxic language**
- Test for **relevance** (chat only)
- Guard against **prompt hacking**

Copilot assembles context from tabs open in the code editor



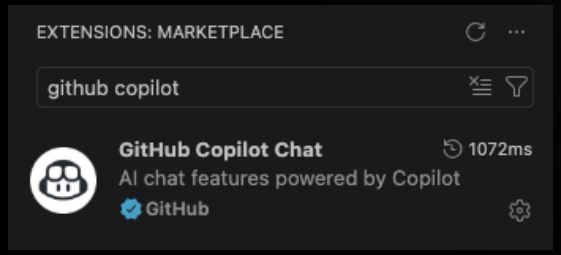
Apply several **post-model filters**

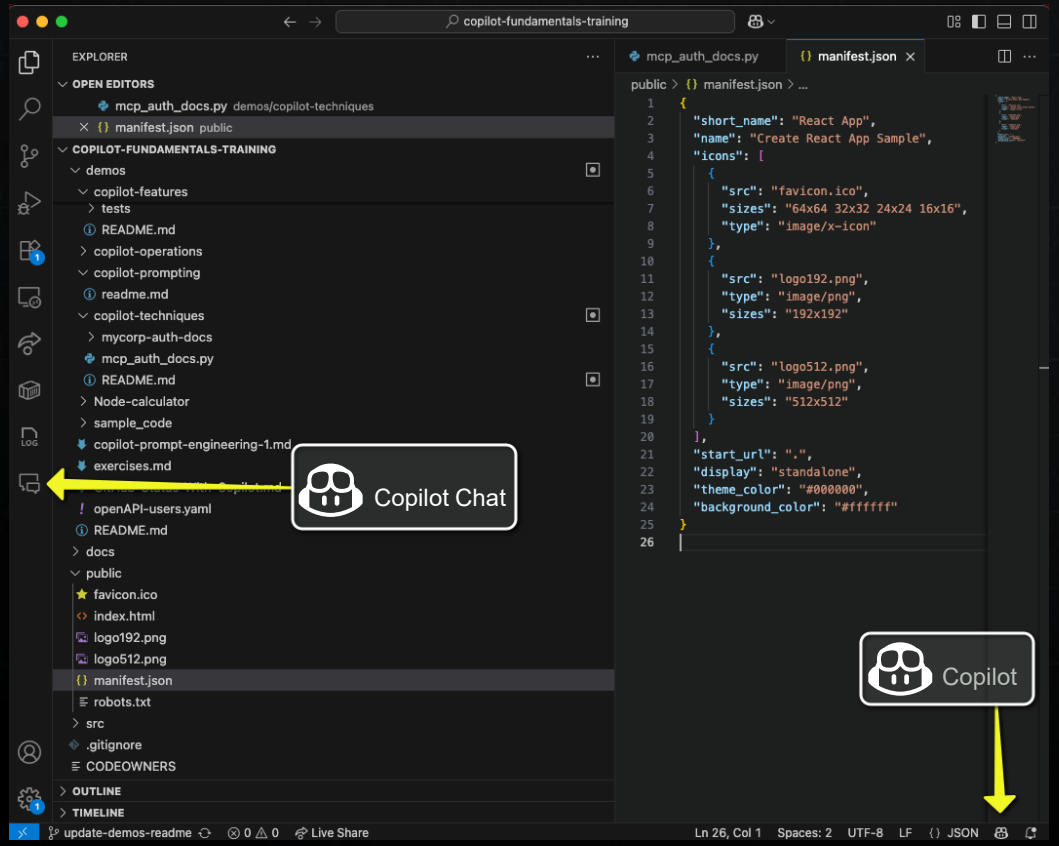
- **Code quality**
- **Unique Identifiers** (eg.: emails, IPs etc.)
- Suggestion **matching public code**



Copilot in your IDE

IDE Extension



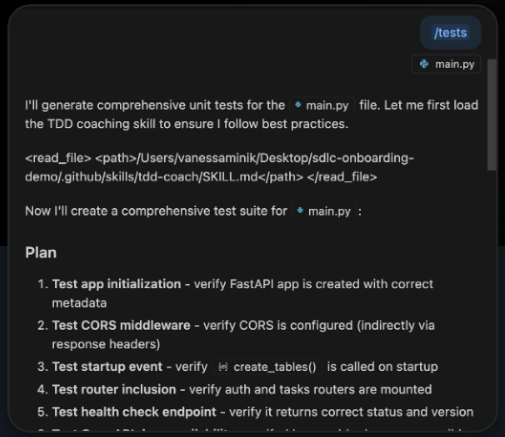


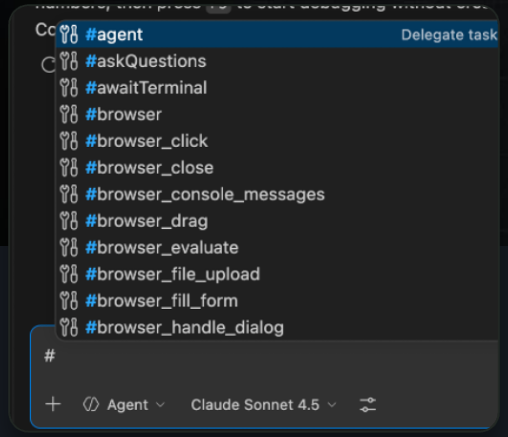
GitHub Copilot fundamentals

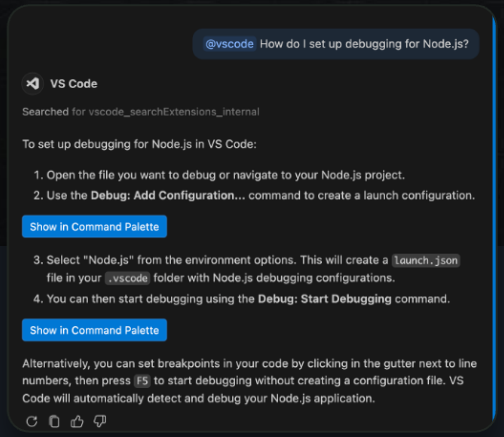
Copilot features



Working with commands







/slash commands

Quick commands

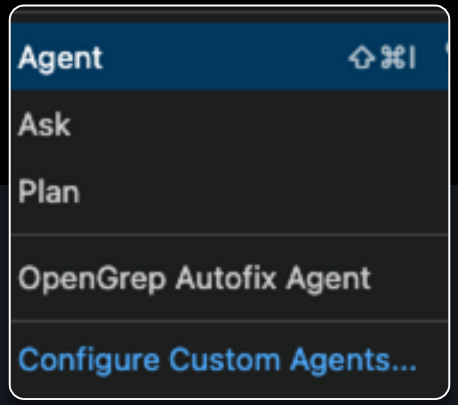
#Chat variables

Add context and tools

@chat participants

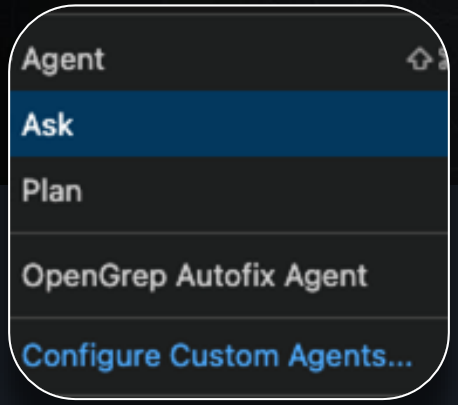
Specify the scope

Chat Overview



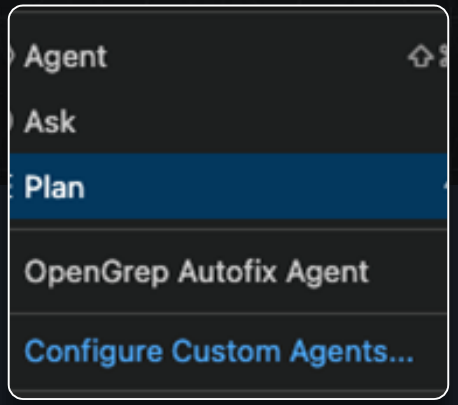
Agent (GA)

Use Copilot Agent to automate repo-wide tasks–refactor, test, commit–from a single prompt.



Ask (GA)

Use GitHub Copilot Ask to generate, explain, or debug code without making changes.



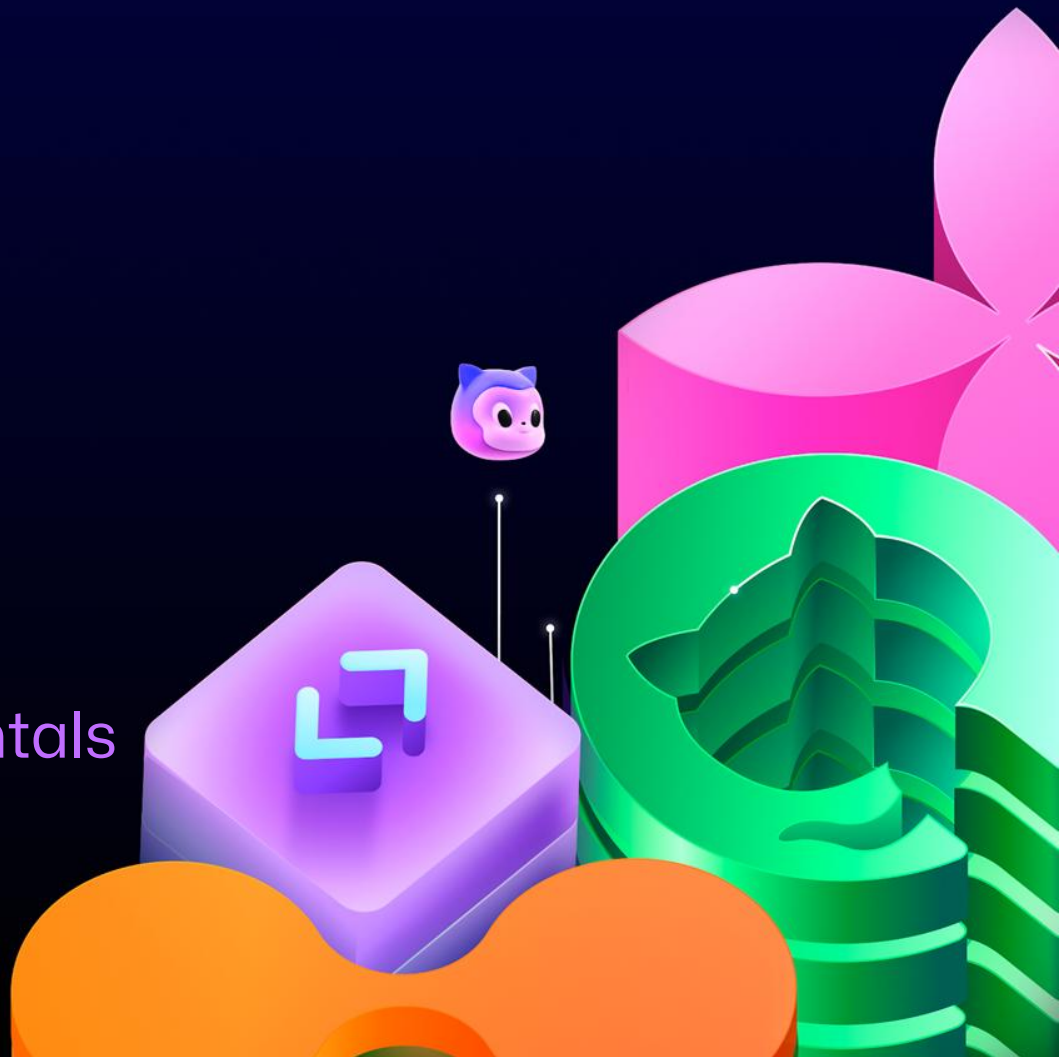
Plan (GA)

Plan mode helps developers create, refine, and execute step-by-step implementation plans directly in their IDE

Demo Copilot features

GitHub Copilot fundamentals

Conclusion



Questions?



GitHub Copilot Session feedback


Thank you